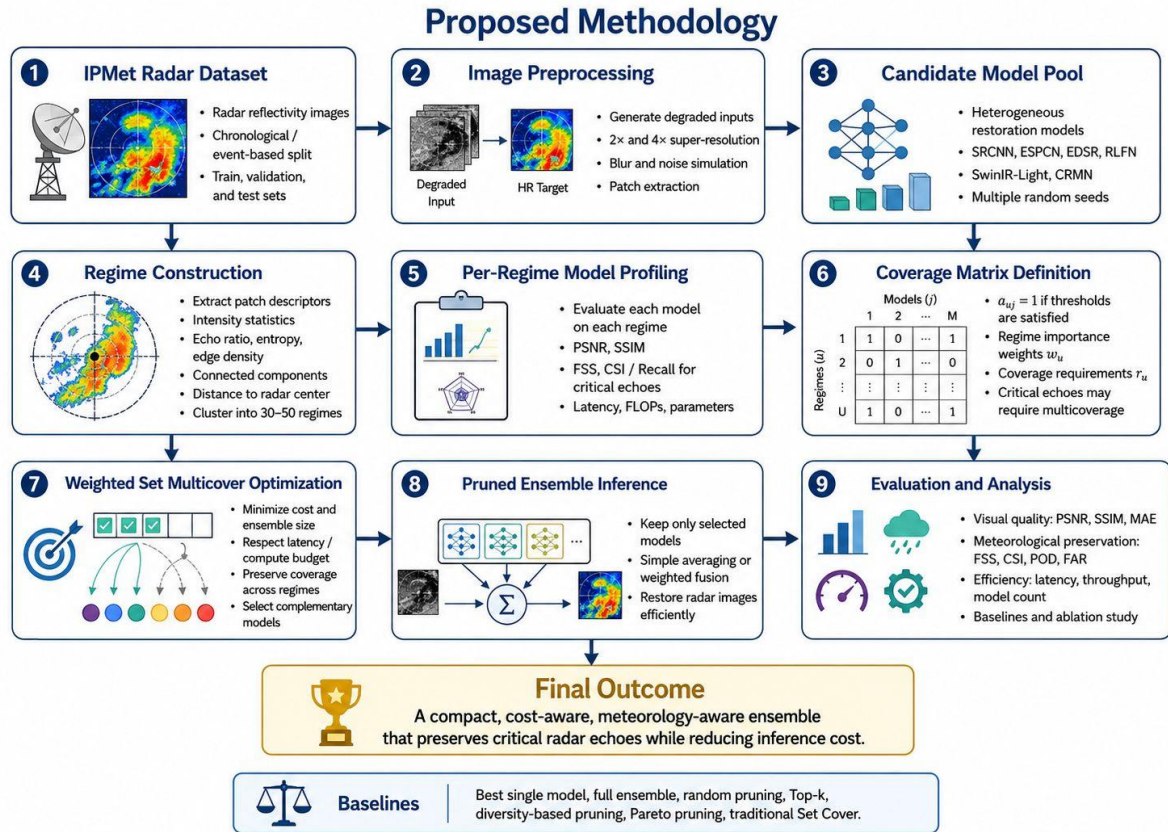




1. Introduction

Present the problem, motivation, research gap, proposed solution, research questions, and main contributions.

2. Related Work

Review radar image restoration, super-resolution models, ensemble learning, ensemble pruning, and Set Covering-based optimisation methods.

3. Proposed Method

Describe the complete RadarCover pipeline, including preprocessing, candidate models, radar regime construction, model profiling, coverage matrix, and Weighted Set Multicover formulation.

4. Experimental Methodology

Describe the IPMet Radar Dataset, chronological or event-based data split, degradation protocol, training settings, baselines, metrics, hardware, and statistical tests.

5. Results and Discussion

Present individual model results, pruning performance, meteorological preservation, computational efficiency, quality–latency trade-offs, qualitative examples, and comparison with baselines.

6. Ablation Study

Evaluate the contribution of each component, such as multicoverage, meteorological metrics, regime weights, computational cost, and weighted fusion.

****7. Limitations****

Discuss limitations related to simulated degradation, threshold selection, clustering quality, dataset generalisation, and training cost.

****8. Conclusion and Future Work****

Summarise the main findings and discuss future extensions, such as dynamic model routing, temporal radar sequences, uncertainty-aware coverage, and additional datasets.

this is the dataset: <https://github.com/rafaepires/IPMet-Radar-Dataset>